



Supporting lower noise in healing environments

Reliable alarm distribution for silent ICU patient rooms

Dräger

Technology for Life

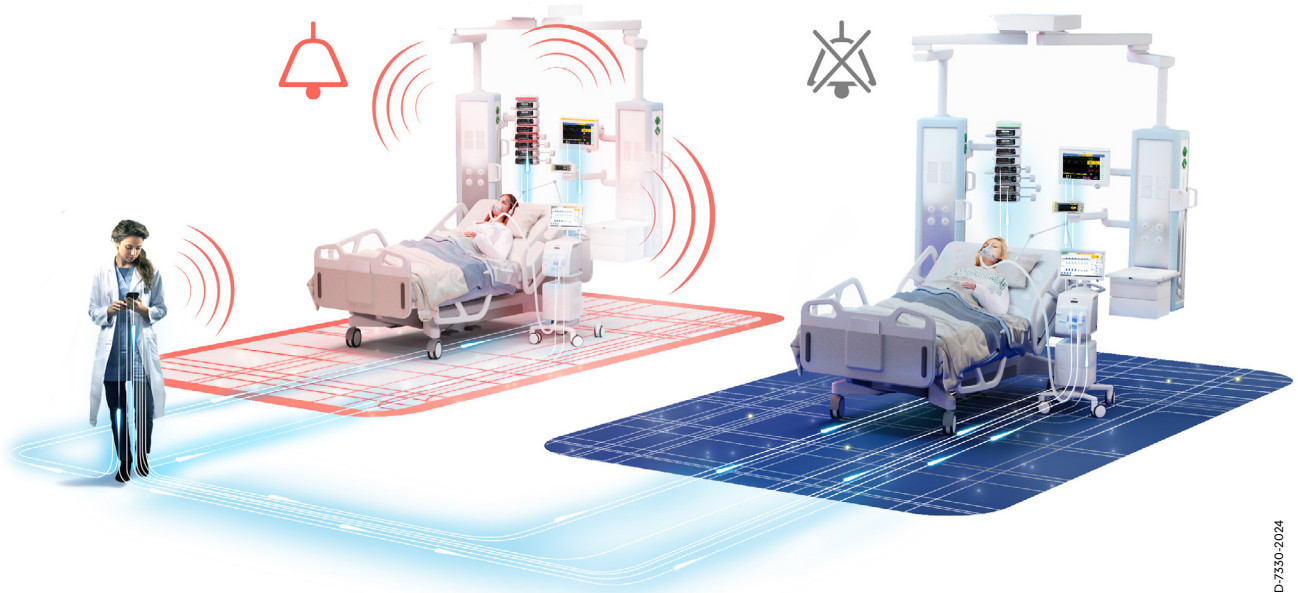


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Consequences of noise pollution on an ICU

Intensive care units are acoustically demanding environments. Medical devices such as patient monitors, ventilators, and infusion pumps can generate up to 350 alarms per ICU bed per day¹, creating avoidable stress for patients. Noise levels may exceed 85 dB several times per hour, while World Health Organization (WHO) guidelines stipulate 35 dB in patient areas.² Excessive noise has been shown to disrupt sleep and recovery and to increase the risk of delirium. Delirium is associated with longer lengths of stay, higher readmission rates, cognitive and functional impairments, and increased mortality.^{3,4}

Studies also indicate that 85–95 % of alarms are non-critical and do not require immediate clinical intervention.^{5,6} Over time, this can lead to alarm fatigue among clinical staff, who may become desensitised to alarms and fail to recognise clinically relevant alerts. Delayed or omitted responses can have serious consequences for patients, highlighting the need to reduce acoustic burden without compromising patient safety.



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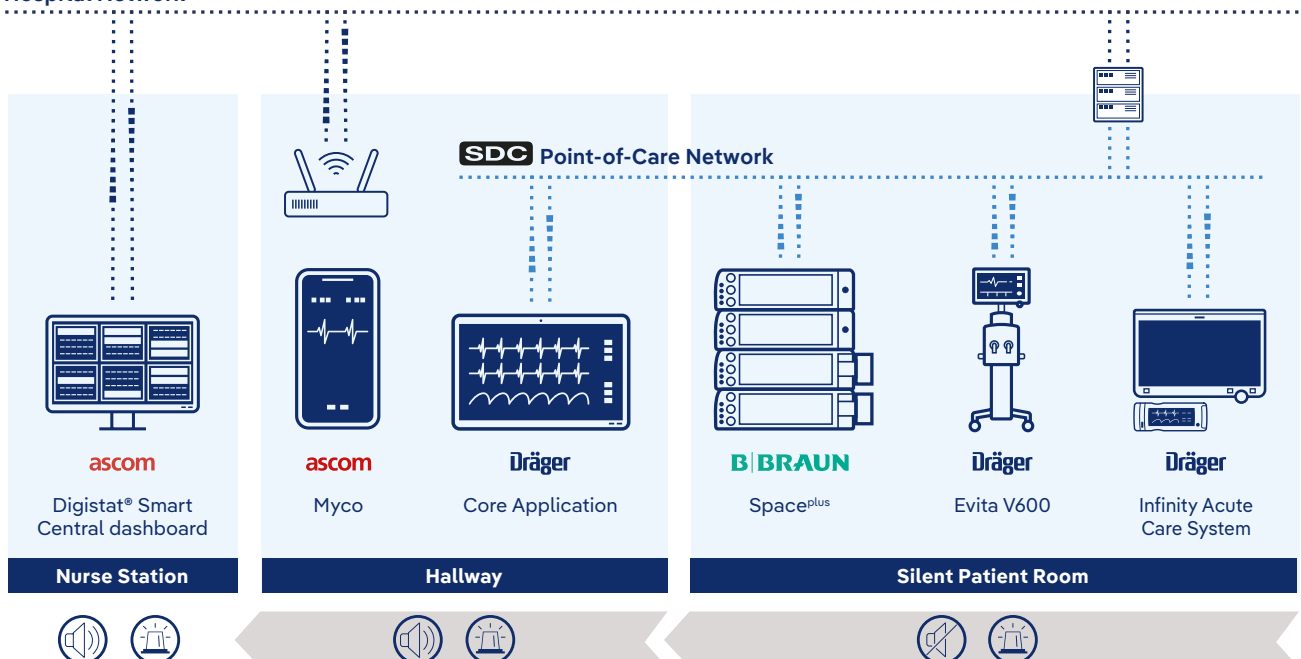
Silent patient rooms through improved alarm handling

Developing smarter solutions to complex clinical challenges requires strong and intelligent connections between industry partners and technologies.

To support a healing environment for patients and more efficient workflows for caregivers, we are collaborating with Ascom and B. Braun to develop a silent ICU. Our joint system solution enables secure, manufacturer-independent distribution of alarms — from bedside devices to a central display and to mobile devices — while bedside alarms remain silent. This can significantly lower noise levels in the patient room and support a healing environment for patients.

The Dräger Silent Care Package is part of one of the first interoperable, multi-modality systems based on SDC technology. The open interoperability standard ISO/IEEE 11073 SDC enables the Dräger Infinity Acute Care System patient monitors, Evita V600 and V800 ventilators, and B. Braun Spaceplus infusion pumps to communicate directly with Ascom's alarm system. Alarms are forwarded directly to the Digistat Smart Central dashboard and the Myco mobile devices used by the responsible nursing staff, ensuring that critical information is received promptly — without unnecessary noise in the patient room. The result is a quieter environment and more efficient alarm handling workflows.

Hospital Network





Assessing impact on outcomes in a clinical study

We are currently evaluating the effectiveness of our Silent ICU solution on intensive care patient outcomes in a clinical study funded by the European Union's Innovative Health Initiative (IHI).^{7,8} The study at the Erasmus Medical Center in Rotterdam, the Netherlands, aims to analyse the impact of quieter patient rooms on patient outcomes and recovery. For this, environmental sound data, alarm logs, clinical data (e.g. delirium scores), and biological data (e.g. stress indicators), as well as physiological and psychological data, are being collected.



Learn more about the
SASICU (Smart and Silent
ICU) project and visit
www.sasicu.eu

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Disclaimer:

Funded by the European Union, the private members, and those contributing partners of the IHI JU. Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the aforementioned parties. Neither of the aforementioned parties can be held responsible for them.



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Secure and reliable alarm distribution powered by open interoperability

The Dräger Silent Care Package, in combination with the Ascom alarm system, functions as a “Distributed Alarm System” (DAS) that meets the requirements of the latest IEC 60601-1-8:2020 standard. A technical feedback channel ensures reliable alarm delivery: the source device receives confirmation that the alarm has been successfully received by the alarm system and delivered to clinical staff. Based on this principle, acoustic alarms in the patient’s room can be silenced while still reliably reaching nursing staff. In the event of a network interruption, a fallback scenario is triggered automatically — including reactivation of bedside acoustic alarms — to maintain patient safety.

References

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